

chromosome. Meiosis and mitosis are both nuclear divisions

that result in new daughter cells. However, the two processes have significant differences. Fill out the following chart comparing the two forms of nuclear division.

	Mitosis (begins with a single cell)	Meiosis (begins with a single cell)
# chromosomes in parent cells		
# DNA replications		
# nuclear divisions		
# daughter cells produced		
purpose		

5. Using your beads, strings, and magnets recreate the process of meiosis. Ensuring you have two different colored beads, demonstrate the process of crossing over. When you think you have it down, flag your instructor over. Have them sign off on your handiwork.
Instructor signature:

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6. By now hopefully you've noticed that these processes are denoted with " $2n$ " and " n " in various places. This is a reference to the number of sets of chromosomes that cell has at any given moment. Autosomal human cells are $2n$. Gametes are $1n$. Mitosis begins with one $2n$ cell and ends with two $2n$ cells. Meiosis begins with one $2n$ cell and ends with 4 $1n$ cells. Sketch those two processes here to show every time the " n " classification changes. (Hint: draw every step, it'll make your life easier, even if it takes a little bit longer!)